

SYSTEM OPERATION AND TECHNICAL DOCUMENTATION

Automated Visual Inspection System for Plastic Parts Using Computer Vision

1. PURPOSE

The purpose of this document is to define the operational procedure and technical overview of the Automated Visual Inspection System developed for detecting color deviations, paint defects, and surface scratches on plastic components.

This document serves as a reference for system operators as well as technical personnel responsible for maintenance and configuration. The objective is to ensure consistent inspection results, reduce manual dependency, and support quality assurance processes within manufacturing environments.

2. SCOPE

This documentation applies to personnel involved in operating and maintaining the visual inspection system, including:

- Machine operators
- Quality control personnel
- Technical supervisors

The document outlines system architecture, hardware and software configuration, operating procedures, troubleshooting methods, and maintenance practices required for reliable system performance.

3. SYSTEM OVERVIEW

The Automated Visual Inspection System is a computer vision–based solution designed to deliver objective, accurate, and repeatable quality inspection for plastic components.

The system utilizes a Raspberry Pi integrated with a Camera Module 3 and a LED lighting setup to capture consistent, high-quality images. Controlled illumination minimizes shadows and reflections, ensuring reliable defect detection.

Captured images are processed through an inspection pipeline that performs:

- Region of Interest (ROI) extraction
- Color deviation analysis
- Scratch detection
- Paint defect identification

Inspection results are displayed through a web-based operator interface, enabling rapid pass/fail decisions while significantly reducing reliance on manual visual inspection.

4. SYSTEM ARCHITECTURE

The system follows a structured workflow where the camera captures images that are processed on the Raspberry Pi. The inspection results are then communicated to the operator through a browser-based interface.

5. HARDWARE CONFIGURATION

The system hardware is designed to maintain imaging consistency and inspection reliability.

Primary Components

- Raspberry Pi (Central Processing Unit)
- Raspberry Pi Camera Module 3
- LED Lighting Assembly for uniform illumination
- Fixed Camera Mount
- Inspection Surface / Fixture

Key Design Considerations

- Stable camera positioning
- Uniform illumination
- Minimal environmental interference
- Controlled capture distance

Multiple lighting configurations were evaluated before finalizing the current setup to achieve optimal inspection consistency

6. SOFTWARE ENVIRONMENT

The inspection system is pre-configured on the Raspberry Pi with all required software dependencies for stable operation.

Core Technologies

- Python-based processing environment
- OpenCV for image processing and defect detection
- Flask framework for hosting the web-based interface
- HTML/CSS/JavaScript for operator dashboard

System configuration and software modifications should only be performed by authorized technical personnel.

7. STANDARD OPERATING PROCEDURE

7.1 System Initialization

1. Power on the Raspberry Pi.
2. Ensure camera connectivity.
3. Switch on LED lighting.
4. Launch the operator interface via web browser.
5. Verify live camera feed.

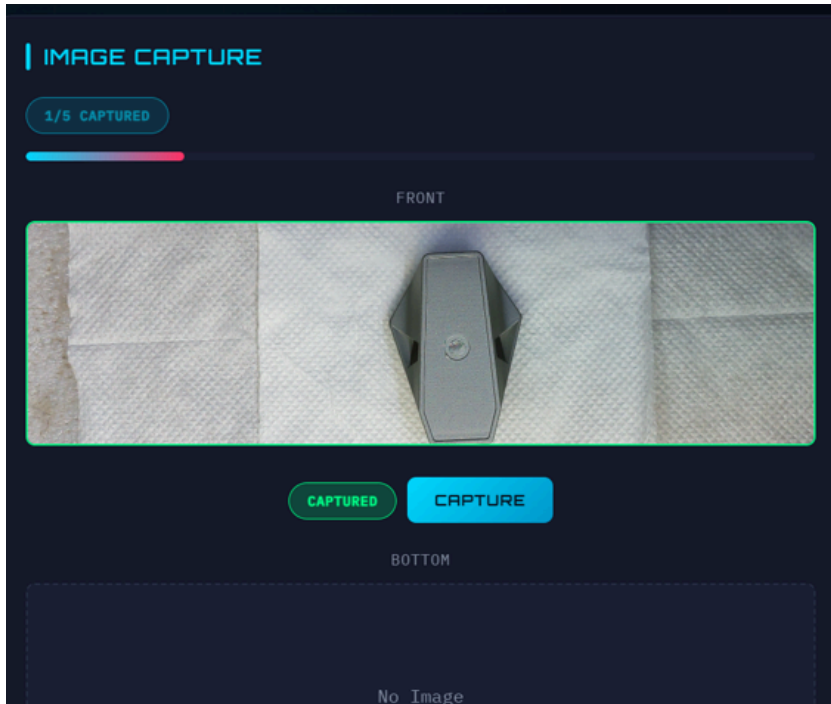
7.2 Component Placement

- Position the plastic component within the inspection area
- Ensure proper alignment

7.3 Image Capture

- Trigger the capture function from the operator interface.
- Confirm that the image is properly illuminated, aligned, and in focus.

Figure 2: Image capture interface.



7.4 Automated Inspection

Start inspection from the dashboard.

The system automatically performs defect detection and classification.

7.5 Result Evaluation

Components are automatically classified as:

- ✓ **PASS** – Meets defined quality standards
- ✗ **FAIL** – Defects detected beyond acceptable tolerance

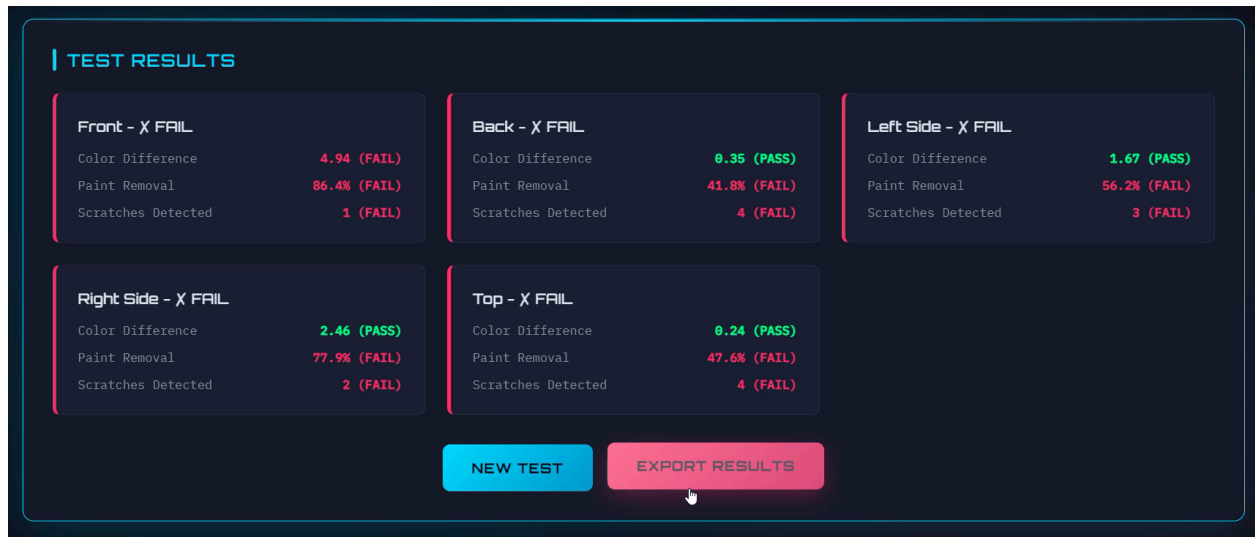


Figure 3: Automated inspection result displaying pass/fail classification.

Rejected components should be removed for further quality review.

8. TROUBLESHOOTING

Image too dark

- Check LED power
- Verify exposure

Blurry image

- Clean lens
- Check mount

False defects

- Revalidate baseline reference images
- Confirm lighting consistency
- Ensure correct component alignment

9. MAINTENANCE GUIDELINES

- Clean the camera lens periodically to prevent image distortion.

- Inspect LED lighting for intensity variation or failure.
- Verify baseline images before initiating production batches.
- Ensure the camera mount remains securely fixed.
- Restart the system periodically for optimal performance.

Signals long-term thinking.

Rare at intern level.

10. TECHNICAL NOTES

- The inspection system operates under controlled lighting conditions to maintain detection accuracy.
- Baseline images are pre-calibrated and should only be updated with engineering approval.
- Threshold parameters are optimized to balance defect sensitivity and false detection rates.

11. SAFETY GUIDELINES

- Avoid moving the camera after system calibration.
- Maintain a stable power supply during operation.
- Keep liquids away from electronic components.
- Operate the system in a controlled lighting environment.

12. SYSTEM ADVANTAGES

- Reduces dependency on manual inspection
- Improves defect detection consistency
- Enables objective pass/fail decisions
- Enhances inspection speed
- Supports scalable deployment in manufacturing workflows

13. ROI RECONFIGURATION PROCEDURE

The Region of Interest (ROI) defines the critical inspection area used by the system for defect detection and color analysis. ROI masks are initially created using baseline reference images.

ROI reconfiguration may be required under the following conditions:

- Installation of a new inspection fixture
- Change in camera position or angle
- Modification in lighting setup
- Introduction of a new device variant
- Replacement of baseline reference images

Failure to redraw the ROI after such changes may result in inaccurate inspection outcomes.

13.1 Prerequisites

Before performing ROI reconfiguration, ensure that:

- The camera is securely mounted in its final position
- Lighting conditions are stable
- The inspection fixture is finalized
- New reference images have been captured

ROI should always be created using the **final production setup**.

13.2 Deleting Existing ROI Masks

1. Navigate to the project directory on the Raspberry Pi or host system.
2. Open the folder:

`roi_masks/shared`
3. Identify the previously generated ROI mask files.
4. Delete the outdated mask images to prevent the system from referencing incorrect inspection regions.

Important:

Ensure that obsolete ROI files are removed before generating new ones. Retaining older masks may cause inspection misalignment.

13.3 Creating New ROI Masks

1. Launch the ROI selection script
2. Load the newly captured reference image.

3. Carefully draw the ROI around the critical inspection region.

NOTE: Prefer drawing the ROI in laptop for accurate region.

Best Practices While Drawing ROI:

- Cover the complete inspection area without including unnecessary background.
- Avoid edges that may introduce lighting variation.
- Maintain consistent margins around the component.
- Ensure symmetry wherever applicable.

4. Save the ROI mask.

The newly generated mask will automatically be stored inside:

`roi_masks/shared`

13.4 ROI Creation for Multiple Devices

If the system supports multiple device variants:

- Repeat the ROI creation process separately for each device.
- Ensure the correct reference image is used for the corresponding device.
- Verify mask naming conventions to avoid file conflicts.

13.5 Validation After ROI Redraw

After generating new ROI masks:

1. Run a test inspection using a known-good component.
2. Confirm that the inspection region aligns correctly.
3. Verify that no background areas are included.
4. Check that defect detection behaves as expected.

Perform at least **3–5 validation runs** before approving the system for production use.

13.6 Authorization

ROI modification should only be performed by trained technical personnel or quality engineers.

Unauthorized changes may impact inspection reliability.

13.7 When NOT to Redraw ROI

ROI should NOT be modified for:

- Minor brightness fluctuations
- Software restarts
- Routine system operation

Only redraw ROI when there is a **physical setup change**.

14. REFERENCE IMAGE CAPTURE PROCEDURE

Reference images serve as the baseline for all inspection activities. Accurate reference image capture is critical to ensure reliable ROI generation and defect detection.

Reference images must always be captured using the finalized production setup.

14.1 Prerequisites

Before capturing reference images, ensure the following:

- The inspection fixture is properly installed inside the jig.
- The component is correctly seated and aligned.
- Camera position is fixed and calibrated.
- Lighting conditions are stable and set to production levels.

Avoid moving the camera or lighting after calibration.

14.2 Fixture Placement

1. Insert the fixture securely inside the inspection jig.
2. Confirm that the device is positioned correctly.
3. Ensure there is no tilt, obstruction, or unwanted background visible.

Proper alignment at this stage prevents ROI misalignment later.

14.3 Capturing Reference Images

Use the following command to capture high-resolution images from the Raspberry Pi camera:

```
rpicam-still --roi 0.25,0.25,0.5,0.5 --width 4680 --height 2592 --awb
daylight --exposure normal --autofocus-mode auto --timeout 5000 -o
<image_name>.jpg
```

Replace `<image_name>` with an appropriate filename representing the device and face orientation.

Example:

`C50_front_reference.jpg`

`C20_front_reference.jpg`

14.4 Image Storage Location

All captured reference images must be stored in the following directory:

`Images/references`

Maintaining a structured folder ensures easier retrieval and prevents incorrect image mapping during inspection.

14.5 Resolution Adjustment (If Required)

The default resolution is configured for high inspection accuracy. However, width and height parameters may be adjusted depending on fixture size and framing requirements.

Ensure that:

- The component occupies a major portion of the frame.
- Minimal background is visible.
- Image clarity is preserved after adjustment.

Avoid frequent resolution changes once production settings are finalized.

14.6 Capturing Multiple Faces

If the device requires inspection from multiple sides:

- Maintain consistent lighting and positioning across captures.

- Use clear naming conventions to prevent confusion.

Example naming format:

device_front.jpg

device_back.jpg

device_left.jpg

device_right.jpg

Consistency in naming helps prevent path mismatches within the inspection pipeline.

14.7 Updating Reference Image Path

After capturing the images:

1. Update the reference image path in the inspection code.
2. Verify that the system loads the correct reference image before proceeding.

Incorrect path configuration may lead to inspection errors.

14.8 ROI Creation

Once the reference images are configured:

- Follow the **ROI Reconfiguration Procedure (Section 13)** to generate new ROI masks.
- Validate alignment before enabling production inspection.

 ROI must always correspond to the latest reference image.

14.9 Validation Before Production Use

Before approving the system:

1. Perform trial inspections using known-good components.
2. Confirm proper ROI alignment.
3. Verify accurate defect detection.

A minimum of **3–5 successful validation runs** is recommended.

NOTE: Reference image calibration directly impacts inspection accuracy and must be performed with precision.

15. CHANGE MANAGEMENT

Any modification to the following requires recalibration:

- Camera position
- Lighting setup
- Fixture design
- Reference images
- ROI masks

All changes should be validated before resuming production inspection.

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